

$$L = T - V = \frac{1}{2} \sum_{\substack{\uparrow \\ w_{ij} = w_{ji}}} w_{ij}(q) q_i q_j - V$$

$\uparrow$ $V \neq V(t)$

stabilna ravn. lega: $\left. \frac{\partial V}{\partial q_i} \right|_{q^0} = 0$

$$\eta_i = q_i - q_i^0 \quad \frac{\partial^2 V}{\partial q_i \partial q_j} > 0$$

$$L = \frac{1}{2} \sum \underbrace{w_{ij}(q^0)}_{T_{ij}} \dot{q}_i \dot{q}_j - \frac{1}{2} \sum \underbrace{\frac{\partial^2 V}{\partial q_i \partial q_j}}_{V_{ij}} \eta_i \eta_j = \frac{1}{2} \dot{\eta}^T \underline{T} \dot{\eta} - \frac{1}{2} \eta^T \underline{V} \eta \Rightarrow \underline{T} \ddot{\eta} + \underline{V} \eta = 0$$

$$\eta_i = \alpha a_i e^{-i\omega t}, \alpha \in \mathbb{C} \rightarrow \underline{V} \underline{a} = \omega^2 \underline{T} \underline{a}$$

$\downarrow$
vektor (lastni)

1) $\underline{T} = T \underline{T} \rightarrow$ iskanje lastnih vrednosti in vektorjev matrice $\underline{V}$

2) $(\underline{V} - \omega^2 \underline{T}) \underline{a} = 0 \rightarrow \det(\underline{V} - \omega^2 \underline{T}) = 0 \rightarrow$ iskanje last. vr. in vekt.

$$T = \frac{1}{2} m \sum_{i=1}^5 \dot{z}_i^2$$

$$V = \sum_{i=1}^4 \frac{1}{2} k \left(\sqrt{l^2 + (z_i - z_5)^2} - l_0 \right)^2 = \sum_{i=1}^4 \frac{1}{2} k \left(l \sqrt{1 + \frac{(z_i - z_5)^2}{l^2}} - l_0 \right)^2$$

$$= \sum_{i=1}^4 \frac{1}{2} k \left(l \left(1 + \frac{1}{2} \frac{(z_i - z_5)^2}{l^2} \right) - l_0 \right)^2 = V_0 + \underbrace{\sum_{i,j} V_{ij} z_i z_j}_{\text{želino}}$$

$$= \sum_{i=1}^4 \frac{1}{2} k \left((l - l_0) + \frac{1}{2} \frac{(z_i - z_5)^2}{l} \right)^2 =$$

$$= \sum_{i=1}^4 \frac{1}{2} k \left((l - l_0)^2 + \frac{(l - l_0)}{l} (z_i - z_5)^2 + \cancel{\frac{1}{4} \frac{(z_i - z_5)^4}{l^2}} \right) = \underbrace{2k(l - l_0)^2}_{V_0} + \sum_{i=1}^4 \frac{1}{2} k \frac{l - l_0}{l} (z_i^2 - 2z_i z_5 + z_5^2) =$$

$$= V_0 + \frac{1}{2} \tilde{k} (z_1^2 + z_2^2 + z_3^2 + z_4^2 + 4z_5^2 - 2z_1 z_5 - 2z_2 z_5 - 2z_3 z_5 - 2z_4 z_5)$$

$\downarrow$
ker more bit simetrična

$$\Rightarrow \underline{V} = \tilde{k} \begin{bmatrix} 1 & 0 & 0 & 0 & -1 \\ 0 & 1 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 1 & -1 \\ -1 & -1 & -1 & -1 & 4 \end{bmatrix}$$

$$\underline{T} = m \underline{I} \rightarrow \underline{V}a = \omega^2 m a \quad \text{lustni vektorji: } a_1 = \begin{pmatrix} 1 \\ 1 \\ -1 \\ -1 \\ 0 \end{pmatrix} \rightarrow \omega_1 = \sqrt{\frac{k}{m}}$$

$$a_4 = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} \rightarrow \underline{V}a_4 = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} \rightarrow \omega_4 = 0$$

$$a_2 = \begin{pmatrix} 1 \\ -1 \\ -1 \\ 1 \\ 0 \end{pmatrix} \rightarrow \omega_2 = \omega_1 = \sqrt{\frac{k}{m}}$$

$$a_5 = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ -\alpha \end{pmatrix} \rightarrow \underline{V}a_5 = \begin{pmatrix} 1+\alpha \\ 1+\alpha \\ 1+\alpha \\ 1+\alpha \\ 4-4\alpha \end{pmatrix} = \lambda a_5 \quad \begin{array}{l} \rightarrow 1+\alpha = \lambda \\ -4(1+\alpha) = -\alpha\lambda \end{array}$$

$$\rightarrow \alpha = 4 \text{ ali } \lambda = 0 \text{ (dobimo } a_4)$$

$$a_3 = \begin{pmatrix} 1 \\ -1 \\ 1 \\ -1 \\ 0 \end{pmatrix} \rightarrow \omega_3 = \omega_1 = \sqrt{\frac{k}{m}}$$

$$\rightarrow a_5 = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ -4 \end{pmatrix} \rightarrow \underline{V}a_5 = \tilde{k} \begin{pmatrix} 5 \\ 5 \\ 5 \\ 5 \\ -20 \end{pmatrix} = 4\tilde{k}a_5 = \omega_5^2 m a_5 \rightarrow \omega_5 = \sqrt{\frac{4\tilde{k}}{m}}$$